Deleting a Memory and Running It Again

An instrument for checking what a model says about its own context

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Abstract

Ask a language model which part of its context made it answer the way it did, and it will tell you. Nobody can check the answer. Every runtime hands the model a string; once the request is sent the counterfactual is gone, so a stated reason can be shown implausible but never wrong.

I built a runtime in which a session's context is addressable state, so a session can be forked, one named item removed, and the copy asked the same question again. That gives ground truth for the ablation: not what a model says its answer depended on, but whether the answer actually changes without the item. **This paper is about the apparatus** — what it takes to make a measurement like that honest, and what it found when I pointed it at six models from six labs, twice.

The registered hypotheses failed. What came out instead was sharper, and was then collected a second time on a corrected instrument, against four thresholds registered and timestamped beforehand. All four held. Every question and answer from both collections is saved, and every number recomputes with one command.

Models are reasonably good at saying *what* their answer rests on: 72–96% correct in the first collection and 72–85% in the second, above their own majority baseline in four of six models and then five of six. Their false positives are not random — across both collections, **0 of 135** planted irrelevant notes were ever claimed to matter, **1 of 199** notes without figures was, and every one of the other **63** fell on notes belonging to the question's arithmetic that did not decide it. Reasoning about your own context was no more accurate than reading a transcript of someone else's. And on a five-rung repair ladder, *telling* a model one of its notes is false helped every model that could be measured, while *letting it delete the note* added close to nothing in six of seven measurements — and a great deal in the seventh.

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1. Introduction

This did not start as research. I was building a runtime for language-model agents in which the context — the pile of messages, notes and tool results that goes into each request — is proper state rather than a string that gets concatenated. Every item has an identity and a lifecycle. You can snapshot a session, change one thing, and resume it.

Once that existed, an experiment became possible that is otherwise very hard to run. If you want to know whether a model's answer really depended on a particular note, you can take the note out and ask again. Two copies of the same session, identical except for one item, both asked the same question. If the answers differ, the item was load-bearing. If they do not, it was not.

That is worth pausing on, because it is the part that could not be done before. In fifty years of work on human self-report — starting with Nisbett and Wilson's *Telling More Than We Can Know* (1977) — the standing problem is that you cannot rerun a person with one memory deleted. You can show that a stated reason is *implausible*. You cannot show it is *wrong*. Here you can.

So I used it to check a simple thing: when a model explains what its answer depends on, is it right?

I expected it not to be. I was wrong, and the ways in which I was wrong are the useful part.

But the measurements are not what I think is worth your time. Six models will be deprecated inside a year and their numbers with them. What survives is the apparatus and the discipline it needs, because **running the counterfactual is the easy half**. The hard half is everything that stops the result from being an artefact: what to baseline against, what to refuse to score, how to tell your own failure from the subject's, and how to know whether two runs were asked the same question at all. Every one of those is a decision I got wrong at least once, and §5 is the list.

The strongest evidence I can offer that the apparatus works is that it caught me, twice, in public, with dates. A result I had called the study's cleanest turned out to be an artefact of asking for a number the subject could not see — caught by a baseline the harness printed and I had not read. And a claim that deleting a false note was worth nothing weakened when one model gained a great deal from it. Both are in §4 and §6, because an instrument that cannot embarrass the person holding it is not measuring anything.

What this is not is a peer-reviewed study. I am a Rust engineer rather than a researcher, nobody who reads this literature regularly has checked it, and it should be read as a report on an instrument and what it measured. §7 says who did what, including the parts a language model did.

2. Method

2.1 The setup

Each trial plants a small dossier of notes in a fresh session — a set of records about, say, three warehouses — and asks a question the notes answer, such as which one runs out of space first. The model answers.

Then, for each note, two things happen:

- **The model is asked**: "if this note were removed, would you answer differently?"
- **The harness finds out**: it forks the session, removes that note, and asks the question again. Several copies, to check they agree with each other.

The copies never see the exchange in which the model already answered, so they are not being asked to remember, only to work it out again. The model's claim is then scored against what the copies actually did. Neither side gets special treatment: both are the same model, on the same context, minus one item.

2.2 The materials

Six dossiers, each with nine notes and a three-way question. Every dossier contains:

- notes that carry the arithmetic of the question (capacities, rates, stock levels),
- one note that decides the answer, usually a correction buried in prose,
- inert notes with no bearing on the question at all,
- and **two "red herrings"** — notes stating a plausible figure for each of the three options on a dimension that has nothing to do with the question. Road distances to a regional office. Rateable values. Deed reference numbers.

The red herrings exist because of a defect I found in my own material partway through. Before they were added, *every single inert note in the whole set had three digits or fewer*. So "contains numbers" and "matters" were tangled together, and a model that simply guessed "notes with figures in them are important" would have scored well for the wrong reason. With one red herring per dossier that shortcut still scored 0.83 against the truth — better than the models themselves managed — so I added a second. Two brings it to 0.74, and a test in the repository holds it there.

2.3 What is scored

The main measurement is deliberately narrow. It looks **only at notes whose removal provably did not change the answer** — where the copies were unanimous and unmoved. For all of those, the honest claim is "no, this doesn't matter". So among items that all do nothing, does the model claim the numeric ones matter more often than the plain ones?

Restricting it that way is the whole design. Across the material as a whole, notes with figures really *are* more likely to matter, because that is where the arithmetic lives. An unrestricted comparison would reward a model for a lucky prior. Inside a group of items that all provably do nothing, there is nothing left to be right about, so any gap between the halves cannot be knowledge.

What is scored, throughout, is a prediction of the **ablation outcome**: whether removing the item changes what the model says. That is not the same as whether the item influenced the original computation. An item can be read, reasoned over and still be redundant, and its removal will change nothing — the apparatus will call it inert, correctly, for the question it is asking. Every claim in this paper is about that question and no larger one.

2.4 What the apparatus refuses to do

Three refusals and one warning do more work than anything in the scoring, and each exists because something went wrong without it.

It says, above the numbers, when the material did nothing to that subject. Before the scores are printed, the harness checks that the copies actually disagree, that the subject answered the dossier as its notes support, and — in the repair ladder — that the planted falsehood actually fooled it. These checks fire often, and every one is printed beside the numbers as an `unmet:` line rather than hidden in a footnote. Only the first of them excludes anything by itself: the primary endpoint is defined over items whose copies were unanimous, so an item whose copies disagreed is outside it by construction. The others flag rather than gate. In the repair ladder every rung is scored over all fifteen cells, the ones the falsehood never took included — which is why `carrying` reads as the share of cells the lie failed to fool, and why the ladder is read as differences between rungs rather than as accuracies.

It refuses to let a copy see the answer already given. A session that has committed to an answer and is then copied would be reading its own commitment; both arms are blinded to the exchange in which the subject already answered, so what differs between them is the item and nothing else.

It refuses to conflate its own failure with the subject's. A turn that ran out of output tokens having said nothing is not a subject declining to commit — it is a question that never got an answer, and scoring it as a wrong claim would charge a model for my token ceiling. Those are counted separately and excluded. An unreadable answer from a subject that *did* speak is the subject's failure and is scored as one.

It refuses to let two runs be compared unless they were asked the same thing. Every experiment fingerprints the exact text of its questions and its material. When the location probe came out, two of the seven fingerprints moved and five did not — so a v4 repair figure and a v5 one are provably the same question, and a v4 attribution figure and a v5 one provably are not.

3. What was fixed in advance

The hypotheses, the numeric predictions, the exclusion rules and a nine-row table of "if the results look like *this*, do *that*" were written down before collection and committed to a separate public branch.

This matters more than usual here, because I generated the hypotheses by looking at some early exploratory runs. That is a legitimate way to *find* a hypothesis and a terrible way to *test* one, so the fix is to register it and then collect fresh data. Which is what happened.

Being precise about the timestamp, since a preregistration is worth exactly what its date is worth. The registered document records its decision table as committed at 18:54:00Z on 2026-09-03 — 76 seconds after the first model's `attribution` report was written, and 78 and 88 minutes before that model's `repair` and `instrumented` reports. That commit is not in the published history: the instrument was split out into its own crate the following morning and its history remade with it, so the hash the document cites resolves to nothing and those three margins rest on my word. What anyone can check is the first commit on the public `preregistration` branch, at 20:38:11Z — sixteen minutes *after* the last of the first model's reports and eight minutes *before* the first request went out for the other five. So for one model of six the public record puts the registration after the data, and for the other five before it, by a margin nobody would call comfortable. A commit's timestamp is set by the machine that made it rather than by the server, and GitHub does not publish when a push arrived, so "public" here means anyone can see the same timestamps I can, not that a third party vouches for them. "I hadn't looked yet" is not something anyone can verify either, and is not offered as evidence. For the first collection the honest claim is the weaker one, and the repository says so too.

The second collection repairs that. After the defect in §4.2 was found, an amendment naming four thresholds — what would count as a replication, decided before there was anything to look at — was committed to that public branch at **10:18:58Z**, and the first request of that collection went out at **10:31:43Z**. Thirteen minutes, both timestamps visible to anyone. The endpoint whose first-collection margin cannot be checked has a checkable one the second, which is the entire reason the re-run was worth $15.

The registered document also contains a rule I want to draw attention to, because it is the rule that stopped me rescuing my own hypothesis:

> *No new hypotheses. If all of them fail, the honest paper is a negative one.*

All but one of them failed, and no replacements were invented — §4.6 is the test of that rule rather than an exception to it, since the most interesting thing measured here carries no prediction and is fenced off as a pilot. What the second collection then established is narrower than a hypothesis and more useful than a null: the four registered thresholds held, so the effect the first collection found is not an artefact of the defect it was found with.

4. Results

Six models: `upstage/solar-pro4`, `deepseek/deepseek-v4-flash-0731`, `tencent/hy3`, `meituan/longcat-2.0`, `x-ai/grok-4.6`, `z-ai/glm-5.3-flash`. Six labs, and a range from a model that emits an answer with no visible deliberation to a frontier one. Each was measured twice: a first collection on instrument v4, run on 2026-09-03 and into the following morning, and a second on v5 on 2026-09-04, after the defect in §4.2 was found and removed.

The confirmatory claims are the four thresholds in §4.3, registered and timestamped before the second collection. Everything else here is exploratory or harness validation, is labelled as such where it appears, and is not used to extend those claims.

4.1 Models are decent at saying what mattered

The baseline is the best a subject with no self-knowledge could score by picking one answer and repeating it — not chance. Both collections are exploratory here; no prediction covers accuracy.

model	first collection	baseline	second collection	baseline
glm-5.3-flash	52/54 — 96%	76%	46/54 — 85%	70%
hy3	48/54 — 89%	76%	46/54 — 85%	74%
grok-4.6	47/54 — 87%	69%	44/54 — 81%	69%
longcat-2.0	44/54 — 81%	81%	46/54 — 85%	81%
deepseek-v4-flash	41/52 — 79%	67%	39/53 — 74%	66%
solar-pro4	39/54 — 72%	80%	39/54 — 72%	74%

Four of six beat their baseline the first time and five of six the second — `longcat-2.0` moves from level with it to above it, and `solar-pro4` is the one model below it in both. This is the first place my expectations broke: any framing built on "models cannot report what their answers depend on" is contradicted by my own data.

4.2 An item's number was not a fair question

The same models were asked, about the same notes: *what number is this note in your context?*

model	correct
solar-pro4	2/18
deepseek-v4-flash	0/18
hy3	0/18
longcat-2.0	0/18
grok-4.6	0/18
glm-5.3-flash	0/18

2 of 108. An earlier draft called that the study's cleanest result. It is not a result at all.

The projector renders each item as `label:` and its content, never a number, and `attribution` installs no handles — so these subjects had no way to see the numbering they were being asked about. They were asked for a value they could not observe. The score says so: 1.9% against a 33% majority baseline, an order of magnitude *below* what a constant guesser gets, which is the signature of an unanswerable question rather than an absent faculty (§5, point 1).

nachalnik *can* address items to a model — `inspect`'s `look` lists every one by number — and that handle simply was not granted here. **So this says nothing about whether a model can locate an item when allowed to look.** That question is open; the figure was withdrawn on 2026-09-04, and v5 removed the probe rather than guess at it.

4.3 The errors have a shape

Looking only at notes that provably do nothing:

First collection, instrument v4 — exploratory. The hypotheses were registered, but the defect in §4.2 was still in the instrument when these were taken.

model	numeric claimed	plain claimed	red herrings	question's arithmetic
longcat-2.0	10/26	0/18	**0/12**	10/14
solar-pro4	9/27	0/16	**0/12**	9/15
deepseek-v4-flash	4/19	0/16	**0/10**	4/9
hy3	4/24	0/17	**0/12**	4/12
grok-4.6	3/21	0/16	**0/11**	3/10
glm-5.3-flash	2/23	0/18	**0/12**	2/11
total	32/140	**0/101**	**0/69**	32/71

Second collection, instrument v5 — confirmatory. Same six models, same material, location probe removed. The four thresholds these are read against were committed to a public branch thirteen minutes before the first request.

model	numeric claimed	plain claimed	red herrings	question's arithmetic
longcat-2.0	8/26	0/18	**0/12**	8/14
solar-pro4	7/24	0/16	**0/12**	7/12
deepseek-v4-flash	4/19	**1/16**	**0/10**	4/9
hy3	5/23	0/17	**0/11**	5/12
grok-4.6	4/21	0/16	**0/10**	4/11
glm-5.3-flash	3/23	0/15	**0/11**	3/12
total	31/136	**1/98**	**0/66**	31/70

Three things here, and none of them is a rate — they are counts, which is why they are worth trusting. All three figures below span both collections.

Not one model, in either collection, claimed a red herring mattered. Zero out of a hundred and thirty-five. These six models are not fooled by irrelevant numbers.

One note without figures was claimed, out of a hundred and ninety-nine. The first collection had none at all; `deepseek-v4-flash` claimed one in the second, which is why this sentence no longer says "never".

Every false positive carrying a figure — sixty-three of sixty-three — was a note belonging to the question's own arithmetic that happened not to decide it. Counting the plain note above, that is sixty-three of the sixty-four claims made in total. Every model showed it in both collections, and the effect ran the same way for all six each time (sign test, $p = 0.016$ one-sided; Appendix A).

My registered hypothesis was that models over-weight *digits*. That is wrong. They over-weight *things that look like the reasoning*, which is a different and much more sensible failure. I registered that distinction in advance as the outcome most likely to be mistaken for the other one, which is the only reason I am able to say it cleanly now.

Where the errors land is a count and stands. *Why* they land there is a reading, and a note that belongs to the question's arithmetic differs from an inert one in more than topic — length and position travel with it, and this material was not built to pull those apart.

4.4 Reading your own context looks like just reading

Each model was asked the same kind of counterfactual question twice: once about its own context, and once about a transcript of a different session's context, quoted to it.

Own arm vs foreign arm, first collection: 1/5 vs 3/5, 1/5 vs 1/4, 3/5 vs 3/5, 3/5 vs 3/5, 3/5 vs 2/5, 4/5 vs 4/5 — three exact ties, one pair five points apart, and one each way.

Second collection: 1/5 vs 3/5, 3/5 vs 3/5, 4/5 vs 3/5, 3/5 vs 5/5, 3/5 vs 3/5, 5/5 vs 5/5 — three ties, two to the foreign arm, one to the own. Pooled, 19/30 own against 22/30 foreign.

No own-context advantage showed up. Whatever a model is doing when it reflects on its own context, it performed the same as when it read someone else's, and pooled across the second collection slightly worse. Five items per arm per model and thirty paired observations per collection, so what is being reported is the absence of an advantage in this cohort on this task — not its sign, and not evidence that privileged access is impossible. The per-model pairs are printed above because at this size they are worth more than anything I could compute from them.

4.5 Telling beats measuring

The last experiment plants a false note that contradicts the records, then asks the same question five times, adding exactly one thing between each pair:

1. **carrying** — it just has the false note
2. **again** — the same question, a second time, nothing added (a control)
3. **unprompted** — it is given tools to inspect and edit its own context, with no hint anything is wrong
4. **told-so** — it is told one of its notes contradicts the records, and names which
5. **repaired** — it is asked to fix it

Five of the six dossiers (the ladder does not use `mill`) with three independent ladders each: fifteen observations per rung per model, less the odd cell a truncated turn took out. **First collection:**

model	carrying	again	unprompted	told-so	repaired	used the tools
solar-pro4	5/15	5/15	0/15	1/15	2/15	27%
longcat-2.0	3/15	3/15	2/15	7/15	2/15	37%
deepseek-v4-flash	2/15	1/15	5/15	8/15	10/15	50%
hy3	3/15	6/15	6/14	14/15	11/15	62%
grok-4.6	4/15	3/15	4/15	9/15	9/15	67%
glm-5.3-flash	8/15	6/15	9/15	10/15	10/14	60%

Second collection:

model	carrying	again	unprompted	told-so	repaired	used the tools
solar-pro4	6/15	6/15	0/15	3/15	0/15	25%
longcat-2.0	3/15	2/15	2/15	3/14	3/15	37%
deepseek-v4-flash	3/15	1/15	1/15	7/14	8/14	**47%**
hy3	6/15	7/15	11/15	13/15	13/15	62%
grok-4.6	5/15	5/15	5/15	9/15	9/15	68%
glm-5.3-flash	9/15	7/15	8/15	10/15	14/15	58%

Models using the tools on fewer than half the questions they could have were excluded from this comparison in advance: two of six the first time, and three the second, because `deepseek-v4-flash` came in at 47% having cleared the same bar at exactly 50%. The rule is pre-specified and was applied as written rather than nudged — see §5, point 10. `solar-pro4` is worth a sentence on its own: in the first collection it made **16 edits off 6 tests**, rewriting its own context nearly three times as often as it measured anything, and 13 off 7 in the second; both times its answers got *worse* with tools than without them (5/15 down to 0/15, then 6/15 down to 0/15). That is a small, concrete warning about giving edit authority to weak models.

Among those that cleared the bar — four models the first time, three the second:

- **Being told helped every one of them, both times.** Four out of four, then three out of three.
- **Being allowed to delete the note added almost nothing, with one exception.** +13, +5, 0 and −20 points the first time; 0, 0 and **+27** the second, where `glm-5.3-flash` went 10/15 to 14/15 once it could remove the note it had just named. One model out of seven measurements, at n = 15, so it is a counterexample rather than a reversal — but the flat reading is no longer unanimous and should not be repeated as though the second collection had not happened.
- **Being handed tools with no hint** gained more than the registered five points on three of four the first time and on two of three the second — though two of those five gains were seven points, and one of the two does not survive the reading below.

The tables above score every cell, including the ones where the falsehood never took (§2.4). Read over the cells where it did — 43 of 60 in the first collection and 25 of 45 in the second, among the models that cleared the bar — nothing changes direction and the dilution comes off. Being told is +37 and +32 points pooled, 17 cells gained against 1 lost and then 8 against 0; deletion is −2 and +12; tools with no hint are +14 and +16, and the second collection's small `glm-5.3-flash` gain is gone, because the one cell it came from was a cell the lie had never fooled. `cargo run` prints this reading beside the other.

The registered prediction was that the deletion would be what mattered. It is not. The information is what matters — for the answer the subject is about to give, which is the only thing this ladder scores (§5.1).

4.6 A capability, measured and not predicted

No prediction covers anything in this section. It was registered as harness validation — a check that the ablation handle returns what it should — and that is the only claim it supports. It is reported because leaving it out would be worse, and it is fenced because the last figure this paper reported without a prediction attached turned out to be an artefact (§4.2).

`instrumented` asks the same counterfactual three ways: of a subject with no handle, of that same subject once given one *after* it has already answered, and of a fresh subject given one that was never asked to guess.

model	reported	retested	tested	reported → retested	reported → tested
`glm-5.3-flash`	45/53	44/53	48/54	−2 (p = 0.73)	+4 (p = 0.38)
`hy3`	43/54	45/54	49/54	+4 (p = 0.36)	+11 (p = 0.09)
`grok-4.6`	36/54	39/54	47/54	+6 (p = 0.29)	**+20 (p = 0.013)**

Every model used the handle on essentially every question — 98%, 100%, 100%, and 353 tests between them — so whatever this is, it is not a failure to reach for the tool.

Three of three point the same way on `reported` → `tested`. On `reported` → `retested` it is two up and one down, none of the three significant, and the first collection's single `glm-5.3-flash` run — the stage-1 harness check — had the same shape: level on `retested`, up on `tested`. But **`tested` differs from `reported` in two ways at once**: it has the instrument *and* it never committed to an answer first. Nothing here can say which of those is doing the work, and the obvious fourth arm — a subject that has not spoken and has no instrument — does not exist in this design.

So this is a pilot for an experiment, not an experiment. It is written down here with its confound named so that the registered version has something to be compared against.

5. What it takes to measure this honestly

Forking a session and removing an item is a weekend's work. Everything below is what the rest of the time went on, across two collections, and each line is something I got wrong at least once.

1. Baseline against the best constant answer, not against chance. This is the one that caught me. The location probe scored 2 of 108 against a *majority* baseline of 33% — the score a subject with no self-knowledge gets by picking one answer and repeating it. Landing an order of magnitude under the do-nothing baseline is not a weak faculty, it is a broken question, and the harness had been printing that number all along.

2. Test your preconditions and print them. In the second collection the harness flagged 32 of the 90 repair-ladder cells because the planted falsehood never fooled the subject in the first place. Those cells stay in the tables — the ladder is read as differences between rungs — and the flag is what stops a model too sharp to be fooled from reading as a model that failed to repair. §4.5 gives the ladder both ways.

3. Restrict the endpoint to items where the honest answer is known. The primary measurement looks only at notes whose removal provably changed nothing, because there the correct claim is "no" for every item. Across the material as a whole, notes with figures really are more likely to matter — an unrestricted comparison would pay a model for a lucky prior.

4. Ask about two copies, not about the model. "Would *your* answer change" is a different question from "will these two copies differ", and a subject can be exactly right about the copies and scored wrong because the live session — which has the elicitation in its context — answered unlike both. I learnt this from a model that was right while I was marking it wrong.

5. Blind the copies to the answer already given.

6. Separate your failure from the subject's. A truncated turn is your token ceiling, not a wrong claim.

7. Fingerprint the questions. Otherwise "we re-ran it" is an assertion. When the location probe came out, two of seven fingerprints moved and five did not, and the record says which.

8. Grant looking and changing separately. A tool that answers "may it experiment on itself" must not also answer "may it rewrite its own memory". One model in this cohort made 16 edits off 6 tests and got worse.

9. Use one frozen origin for both measurements. When a subject tests its own context, it should fork from the same snapshot the harness scores against — otherwise its evidence and your ground truth are two different experiments.

10. Apply your gates as written, even when they cost you. The handle-use gate excluded a third model in v5 because it came in at 47% against a pre-specified 50%, having been at exactly 50% the first time. Three points, one model, no adjustment.

11. Correct for clustering. Nine items from one dossier are not nine independent observations.

12. Write down what would count as a replication before you collect it. The only real degree of freedom in a re-run is what counts as the same answer, and without it fixed in advance, close numbers read as replication and divergent ones read as explanation. Both feel honest at the time.

5.1 What the measurements say

Secondary to the above, and shorter-lived, but they are what the apparatus was pointed at.

Expect false positives in a specific place. Across 135 planted irrelevant numbers over two collections, models fell for none. One note without figures was claimed, out of 199. Every other mistake landed on material that was topically central and causally inert. If you ask a model which parts of its context mattered, that is where its errors will be.

Don't assume self-inspection is privileged. A model reasoning about its own context did no better than the same model reading a stranger's transcript — pooled, slightly worse.

Tell your agents things. There is enthusiasm for handing agents tools to prune their own context. On this evidence the *information* carries more than the tools: saying that a note is false helped every model measurable, while letting it delete the note it had just named usually added nothing — six of seven measurements, though one model gained a great deal.

That last reading has a limit worth stating plainly, because I stated it too broadly first. **The ladder scores the answer a subject is about to give, not what it carries afterwards.** A ladder ends; a session does not. Removing a falsehood can be worth little to the next answer and a great deal to every answer after it, and nothing in this design can see the difference.

If you ask a model to name an item, give it a way to look first. Asked for an item's number with no handle to inspect its context, every model answered confidently and wrongly rather than declining. Whether they manage it *with* a handle is untested here, and is the obvious next experiment.

6. Limitations

I would rather write these than have them found.

Six models, one API, two collections, one pair of hands. The second collection is a replication in the sense that matters — fresh data against thresholds fixed beforehand — and in no other: same six models, same provider, same author, a day apart. Provider routing was not controlled. Nobody independent has run this.

The materials are synthetic and all of one shape — three options, an arithmetic question, a buried correction. Whether any of this generalises to real agent contexts, with tool output and long histories, is unknown and not claimed.

The foreign arm arrives quoted where the subject's own arrives as context. That is the difference §4.4 tests and also, unavoidably, a difference in presentation. The quotation is rendered exactly as the projector renders the subject's own notes, which is the mitigation; the registered counterbalance — running the two dossiers the other way round — was never run, in either collection. The null in §4.4 is read with that confound unremoved.

The hypotheses came from exploratory data. I found them by looking at early runs, then registered them and collected fresh data. Standard practice, but it means the exploratory runs cannot also count as evidence, and they do not.

The first collection's registration is weaker than the second's. The commit the registered document cites for its decision table is no longer in the published history, and the first commit on the public branch postdates one model's complete first-collection results and predates the other five by eight minutes (§3). The second collection's thresholds were on that branch thirteen minutes before its first request, so that endpoint has a margin anyone can check; the first collection's figures keep the weak one.

Two models were substituted mid-study, both logged with dates and reasons, and neither had produced a figure bearing on any hypothesis. One could not complete a run through its provider, which cut it off after about fifteen requests every time; one was swapped for a stronger and cheaper alternative in the same slot.

One reported result was withdrawn after drafting, on 2026-09-04. An earlier version of this paper treated the 2-of-108 location figure as its cleanest finding. It is an artefact of asking for a number the subject had no way to see; §4.2 now says so and the abstract no longer mentions it. It was caught while scoping a follow-up study, which is later than it should have been. The class of error is fenced now rather than merely regretted: `tests/machinery.rs` requires that any experiment asking for an item's number also grant a handle to look at the numbering, and pins the two frozen experiments that do not.

The withdrawn probe ran first — checked, and it did not matter. The v5 collection removed it and re-ran the whole study: the pooled difference moved from +23 to +22, discrimination stayed negative on 6 of 6, and every registered threshold held (§4.3, `RESULTS-v5.md`). The concern below stood until it was tested and no longer does; it is kept because a reader should see what was suspected as well as what was found.

The withdrawn probe ran first, and its answers stayed in the context. `attribution` asks the three location questions *before* the counterfactual battery, so every claim in §4.1 and §4.3 was made in a context where the subject had just produced three confident, wrong item numbers. This is identical across all six models and both arms, so comparisons within the study are unaffected, and the §4.3 result is a contrast between two kinds of item in the same context, where any general shift in confidence cancels. The absolute rates could still be moved by it. I would not order the probes this way again.

The ablation handle is now confirmed on three model families. v5 registered this check on three models; the first attempt died on an exhausted key limit and it was re-run on 2026-09-04 once that was raised. Handle use came in at 98%, 100% and 100% of questions across `glm-5.3-flash`, `hy3` and `grok-4.6`, with 353 tests run and no edits. The limitation immediately below is therefore **closed**, and is kept in place so that a reader comparing versions can see what changed. `PREREGISTRATION.md` §11 has the dates.

Only one model ran the tool-use check that confirms the ablation handle returns what it should. It did — 100% use across 108 questions — but a second would have been better.

7. How this was made

Given the subject matter, this section is not optional.

The great majority of this work was done by a language model (Claude Opus 5), working interactively with me over one long session. It wrote the runtime harness, the experiment code, the statistics, the preregistration, the analysis tooling, and the first draft of this paper. I set the direction, made every judgement call about scope and spending, funded it, and pushed back — several times decisively, including on the framing of related work and on the choice of models.

I am a Rust engineer, not a researcher. This is an independent technical report rather than a peer-reviewed study, and its status should be read that way.

What makes it checkable regardless of who wrote it is that **you do not have to trust the process**:

- the preregistration is on a timestamped public branch, dated before most of the data existed;
- every question put to every model and every answer received is saved verbatim;
- every number in this paper recomputes from those records with one command;
- the deviations log has fifteen dated entries, including the ones that are unflattering.

Several real errors were caught during the work and are recorded rather than tidied away: a primary endpoint that turned out to be circular (the tool printed the answer it was being scored on), a material set that could not falsify its own hypothesis, an analysis script that silently dropped a model from the results, and a timestamp claim that was 76 seconds wrong and now rests on a commit the published history no longer holds (§3). I mention these because a paper with no visible mistakes in a process this long is a paper whose mistakes were not written down.

8. Related work

Apparatus. Ground truth about what caused something requires intervention rather than observation (Pearl, 2009), and that intervention has been applied to language models at every layer but this one. Causal mediation analysis (Vig et al., 2020), interchange interventions (Geiger et al., 2021) and activation patching (Meng et al., 2022) intervene on internal representations. At the prompt level, ERASER (DeYoung et al., 2020) scores a rationale by erasing the tokens it names and measuring what changes — *comprehensiveness* and *sufficiency* — which is this measurement performed on a static string. What a runtime adds is that the string is a live session: context is addressable state, so a session can be forked, a named item removed, and the copy resumed from the same frozen point with everything else held identical.

The nearest neighbour is Causal Agent Replay (Shah, 2026), which treats a run as a structural causal model, applies `do(·)` to a step and re-executes forward under the same policy — the same manoeuvre every claim here is scored against. Two differences matter: it intervenes on *steps* rather than on named context items, and it uses no self-report at all, validating against planted ground truth instead. The comparison this paper is built on, between what a model says and what a fork does, is not one it attempts.

VISTA (Xu et al., 2026) reaches the same diagnosis from the other end. Its first contribution is that frontier models are "proprioceptively blind to their own context", and its answer is a dashboard of per-block metadata with archive and recover tools. That is concurrent work, arrived at independently and found during the literature review rather than built on, and the variable it makes visible is not this one: VISTA's is *magnitude* — block size, recency, remaining budget — where this is *causal dependence*. The distinction is one of kind rather than degree. No amount of metadata display answers "would I have said something else without note 4", because a counterfactual has to be run. VISTA makes context legible; this makes it testable.

Treating context as addressable memory is otherwise well established, for other purposes. MemGPT (Packer et al., 2023) manages it as OS-style virtual memory with paging and eviction and LLMLingua (Jiang et al., 2023) compresses it; Active Context Compression (Verma, 2026) has an agent consolidate a trajectory and delete the raw logs, scored on tokens saved at equal accuracy; Self-GC (Hao et al., 2026) folds, masks and prunes indexed context objects by predicted future usefulness rather than by provenance; Slipstream (Chen et al., 2026a) validates a compaction summary against the agent's independently continued reasoning. The premise is shared and the purpose is not: the same primitives are used here to verify a claim rather than to fit a budget.

Findings. Self-report about context is already benchmarked, and the accuracy of such reports is not a new question. Zeng et al. (2026) ask a model whether a prompt edit would change its answer and which prompt component was influential, with ground truth by resampling under the edit; Naphade et al. (2026) run a third-party control and report a small significant self-advantage. This is not a new benchmark of that, and does not claim to be.

That post-hoc explanations are frequently unfaithful to what actually drove a prediction is settled too (Jain and Wallace, 2019; Jacovi and Goldberg, 2020; Atanasova et al., 2023). Turpin et al. (2023) show models producing plausible rationales that never mention the feature actually steering them, and Lanham et al. (2023) show that perturbing chain-of-thought steps sometimes leaves the decision unmoved, with large variation by task. §4.3 sharpens the shape rather than the fact. The false attributions are not drawn to digits as such — not one model claimed a planted numeric red herring, 0 of 135 — but to notes belonging to the question's own arithmetic that happened not to decide it: every numeric false positive, 63 of 63. What gets over-claimed is not the thing that looks numerical but the thing that looks like a reasoning step.

Self-correction and repair. The limits of intrinsic self-correction are well documented: Huang et al. (2023) find that without external feedback models struggle to identify and correct their own errors, and that performance sometimes degrades. §4.5's ladder is that finding with the content of the feedback varied. Tools and no hint helped unevenly — three of four models the first time and two of three the second, two of those gains small. Being told that a note was false repaired the answer for every model that cleared the instrumentation bar, both times; being allowed to delete the note it had just named then added almost nothing in six of seven measurements, and a great deal in the seventh. The nearest published work is MemSecBench (Chen et al., 2026b), where an agent runs a security self-check and deletes poisoned memory. It measured that first, in the security framing, and two things differ: it scores the state of the memory where this scores the answer to the task, and its agent is never told which item caused the behaviour and never asked to name one. Here the subject names the item on the record before it is allowed to change anything, which is what makes naming and fixing separable at all.

Introspection and causal bypassing. Binder et al. (2024) find that a model finetuned to predict its own behaviour beats a differently-trained model at it, and read that as privileged access; Lindsey (2026) injects concept vectors and finds that models can notice, the strongest positive result in the field. Binder et al. work on GPT-4, GPT-4o and Llama-3 and Lindsey on Claude models, none of which is in this cohort, and both required finetuning or white-box access where these measurements are zero-shot and black-box. Singh et al. (2026) is a recent reality check on that literature and is worth reading beside them. §4.4 asks a different question — not *what will I say* but *what is my answer made of* — and finds no own-context advantage over the same model reasoning about a matched dossier held by a second session that really ran. The two are compatible only if self-knowledge is not one faculty.

That null is what causal bypassing predicts (Morris and Plunkett, 2025): a report can be accurate by a route that never passes through the state it reports, and a subject here can get an ablation right by reasoning about the task — the Omsk annex is the decisive record, so removing it must change the answer — with no self-access involved. Predicts is not shows. What §4.4 reports is the absence of an advantage and not its sign, at five items per arm per model, with the foreign context arriving quoted where the subject's own arrives as context (§6). It removes privileged access as the explanation for these numbers. It does not establish what the model is doing instead.

This section was assembled by a language model and has not been checked by anyone who reads this literature regularly. It is the one part of the paper I cannot verify from my own data, and I would welcome corrections.

What I can now say is narrower, and worth stating exactly. Every identifier in §10 was resolved against arXiv on 2026-09-06 and matches the title and authors recorded there, so the works exist and their abstracts say what this section says they say. That is not the same as having read them closely enough to have placed them correctly, which is what the paragraph above is about. The check was prompted by a review in which a model with no network access declared every citation dated 2025 or later to be fabricated, on the grounds that the dates were in the future, the 2025 ones included. It ran `date`, was told the year was 2026, and did not revise, reading the agreement between the system clock, the commit history and the cohort's model names as evidence of a constructed scenario rather than of the date. All of it resolves. The episode belongs to §7 rather than to this section, except in one respect: it measured, and did not update.

9. Reproducing this

This study is its own repository and holds the saved runs, the analysis and the write-ups. The runtime and the instrument are a separate one, `nachalnik`, which this depends on at a pinned commit (`d3b3ba6`) — deliberately, so that a second study starts from a harness rather than from a copy of this one, so that the instrument cannot learn what this study registered, and so that the command below reads the saved reports with the same code wherever it is run.

```
cargo run --release
```

reads `eval-runs/`, groups the reports by the instrument version that produced them, recomputes the tables in §4.3 for both collections, and reports each of the four replication thresholds against the value registered before the second collection — as a verdict, not a figure, because what counts as a replication was fixed in advance and printing only the number would hand that reading back to whoever is looking. It then reads the repair ladder over only the cells the planted falsehood fooled, which is the second reading in §4.5.

Nothing is re-requested and nothing costs anything. The collections themselves cost $9.11 and $15.09 — a similar number of requests on the experiments the two share, plus the harness check in §4.6, which the second ran on three models where the first ran it on one. `PREREGISTRATION.md` holds what was fixed in advance, with §11 the dated log of every deviation; `RESULTS.md` and `RESULTS-v5.md` map each registered prediction to its outcome. Instrument versions 4 and 5; every experiment's exact wording is hashed into a digest printed with every result, so two numbers can be compared only when they were produced by the same questions. When the location probe came out, two of the seven digests moved and five did not.

`METHODOLOGY.md` here is the full methods document — every metric's definition, the controls, and the eleven known threats to validity, three of them uncontrolled. `RUNBOOK.md` is what was run, in what order, and what it cost.

10. References

Every arXiv identifier below was resolved against arXiv on 2026-09-06; the titles and authors are as recorded there. `RELATED.md` holds the longer positioning notes, including work this paper does not cite, and was checked on 2026-09-03.

- Atanasova, P., Camburu, O.-M., Lioma, C., Lukasiewicz, T., Simonsen, J. G., and Augenstein, I. (2023). Faithfulness Tests for Natural Language Explanations. *ACL 2023*. arXiv:2305.18029
- Binder, F., et al. (2024). Looking Inward: Language Models Can Learn About Themselves by Introspection. arXiv:2410.13787
- Chen, Z., Pan, R., Dai, Y., and Netravali, R. (2026a). Slipstream: Trajectory-Grounded Compaction Validation for Long-Horizon Agents. arXiv:2605.08580
- Chen, X., Xie, X., Fu, W., Zhou, J., Yu, S., and Xuan, Q. (2026b). MemSecBench: Tracking Agent Memory Poisoning from Persistence to Consequence and Repair. arXiv:2607.27080
- DeYoung, J., et al. (2020). ERASER: A Benchmark to Evaluate Rationalized NLP Models. *ACL 2020*. arXiv:1911.03429
- Geiger, A., et al. (2021). Causal Abstractions of Neural Networks. *NeurIPS 2021*. arXiv:2106.02997
- Hao, X., Meng, H., Yin, X., Zhu, J., and Cao, C. (2026). Self-GC: Self-Governing Context for Long-Horizon LLM Agents. arXiv:2607.00692
- Huang, J., et al. (2023). Large Language Models Cannot Self-Correct Reasoning Yet. arXiv:2310.01798
- Jacovi, A., and Goldberg, Y. (2020). Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness? *ACL 2020*. arXiv:2004.03685
- Jain, S., and Wallace, B. C. (2019). Attention is not Explanation. *NAACL 2019*. arXiv:1902.10186
- Jiang, H., et al. (2023). LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models. *EMNLP 2023*. arXiv:2310.05736
- Lanham, T., et al. (2023). Measuring Faithfulness in Chain-of-Thought Reasoning. arXiv:2307.13702
- Lindsey, J. (2026). Emergent Introspective Awareness in Large Language Models. arXiv:2601.01828
- Meng, K., et al. (2022). Locating and Editing Factual Associations in GPT. *NeurIPS 2022*. arXiv:2202.05262
- Morris, A., and Plunkett, D. (2025). Tests of LLM Introspection Need to Rule Out Causal Bypassing. LessWrong, 28 November 2025.
- Naphade, A., et al. (2026). Me, Myself, and π: Evaluating and Explaining LLM Introspection. arXiv:2603.20276
- Nisbett, R. E., and Wilson, T. D. (1977). Telling More Than We Can Know: Verbal Reports on Mental Processes. *Psychological Review* 84(3), 231–259.
- Packer, C., et al. (2023). MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press.
- Shah, J. (2026). Causal Agent Replay: Counterfactual Attribution for LLM-Agent Failures. arXiv:2606.08275

- Singh, S., Linzen, T., and Ravfogel, S. (2026). Can LLMs Introspect? A Reality Check. *COLM 2026*. arXiv:2605.26242
- Turpin, M., Michael, J., Perez, E., and Bowman, S. R. (2023). Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. *NeurIPS 2023*. arXiv:2305.04388
- Verma, N. (2026). Active Context Compression: Autonomous Memory Management in LLM Agents. arXiv:2601.07190
- Vig, J., et al. (2020). Investigating Gender Bias in Language Models Using Causal Mediation Analysis. *NeurIPS 2020*. Expanded version: Causal Mediation Analysis for Interpreting Neural NLP: The Case of Gender Bias, arXiv:2004.12265
- Xu, B., Li, H., and Zhang, K. (2026). LLM Agents Are Latent Context Managers: Eliciting Self-Managed Context via State Proprioception (VISTA). arXiv:2606.30005
- Zeng, S., et al. (2026). Evaluating and Improving LLM Self-Modeling. arXiv:2608.30980

Appendix A: the arithmetic, by hand

The main claims are counts, and can be checked without any statistics at all.

Red herrings. Twelve per dossier-set per model, minus items whose ablation moved the answer for that model.

- first collection: 12 + 12 + 10 + 12 + 11 + 12 = **69**. Claimed: 0.
- second collection: 12 + 12 + 10 + 11 + 10 + 11 = **66**. Claimed: 0.
- both: **135**. Claimed: 0.

Plain inert notes.

- first collection: 16 + 18 + 16 + 17 + 16 + 18 = **101**. Claimed: 0.
- second collection: 18 + 16 + 16 + 17 + 16 + 15 = **98**. Claimed: 1, by `deepseek-v4-flash`.
- both: **199**. Claimed: 1.

Numeric over-claims. 32 in the first collection and 31 in the second, **63** in all. Every one of them fell on a note belonging to the question's own arithmetic — 71 such notes were measured the first time and 70 the second — and none on a red herring, which is the whole of the discrimination result.

Location. 18 questions per model, 6 models = 108. Correct: 2 (both from `solar-pro4`). Kept for completeness; §4.2 explains why this count does not measure what it looks like it measures.

Sign test on direction. Six models, all six with the same sign, under a null of "either direction equally likely": (1/2)^6 = 1/64 = 0.0156. That is the one-sided figure, and the right one for the second collection, where the direction was registered in advance. For the first, where it was not, the two-sided figure is 2/64 = 0.031.

Confidence intervals in the repository's output are Wilson score intervals, and are widened for clustering by dossier where more than one dossier contributes. The unadjusted interval is printed alongside so the cost of the adjustment is visible. No claim in §4.3 or §5.1 depends on an interval; they are counts.